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A robust approach to calculating bridge displacements from unfiltered accelerations for highway and railway bridges

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A R T I C L E I N F O

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A B S T R A C T

Bridge displacement, particularly due to specific loading events, is increasingly seen as a data signal that potentially contains information useful for bridge management. The lack of a convenient fixed reference location historically made displacement measurement challenging in field conditions, but recent developments in image processing have enabled low-cost, camera-based displacement measurements. Even so, logistical challenges remain. These include organising access to locate the camera, stabilising the camera in windy conditions and managing the large data files that result. Consequently, since accelerometers are widely used, research has continued to investigate recovery of displacement through double integration of acceleration signals. The main challenge with this approach is the errors in the calculated displacement signal due to the low frequency noise. Much of the research to overcome this challenge has involved the use of filters, but results can be highly sensitive to parameter selection and it can be difficult to define the confidence level for the calculated displacement. In this paper, an approach is proposed that does not require the use of filters. The approach, instead, focuses on the quality of hardware used to minimise low frequency noise in acceleration signals, and then implements various quality checks for different bridge and loading scenarios. Sensitivity of the proposed procedure to the choice of start and end points in the acceleration time series data is demonstrated and an approach to overcome this sensitivity has been developed. The robustness of the approach was trialled on three different bridges, subject to three different loading scenarios, that is, a moving truck, a moving train, and pseudo static loading with a truck stopping on a bridge. Where the calculated displacements passed quality checks, they were found to be in good agreement with the directly measured displacements (recorded with Imetrum displacement measurement system). Unsurprisingly, long duration, pseudo-static loading is the most challenging case due to cumulation of low frequency noise effects, where the quality checks correctly identified that the calculated signal would be unreliable.

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1. Introduction

Visual inspections of bridges are subjective and leave a wide scope of variability that comes down to the judgement of the individual inspector [1], yet they are still relied on for bridge condition assessments. In the evaluation of 101 instrumented bridges, [2,3], 82 bridges were tested for optimal load carrying capacity. 66 bridges (75 %) were found to have higher load carrying capacities using data driven assessments, such as load tests, compared to analytical evaluations which led to restrictions imposed on the bridges due to visual defects. Whilst there is real motivation for capturing useful data for bridges [4], there are considerable challenges due to scale and logistics. This is particularly true for data such as displacement, which has been shown to be useful in bridge condition assessments, [5], but is difficult to measure using traditional approaches.

1.1. Previous work on identifying bridge displacement

Bridge displacement can be challenging to measure using traditional approaches that rely on fixed reference points which, in many practical cases at a bridge, may not be readily available. State of the art approaches to measuring bridge displacement that exist in the bridge SHM community now include the use of vision-based systems e.g. [5,6]. Whilst these systems can achieve a high level of accuracy, they can also be compromised by environmental effects, such as light levels, camera shake (from wind) and visual obstructions [7–9]. As bridges are rarely sheltered from such effects, it can be challenging to rely on this as a singular approach to measuring bridge displacement.

Another promising approach to measuring bridge displacements is through numerical integration of acceleration signals. The underlying principle adopted here is that 0 mm/s^2 acceleration, 0 mm/s velocity and 0 mm displacement can be assumed for a stationary, unloaded bridge, and this is used to detrend acceleration signals that inherently measure effects of gravity even when stationary. The predominant challenge in the case of vehicle-induced acceleration response is corruption of the acceleration signal by low frequency noise that results in erroneous calculated displacement. To overcome this, the majority of research in this area has studied application of filtering techniques (Section 1.2), with the minority studying noise minimisation approaches (Section 1.3).

1.2. Displacement estimation from filtered acceleration

Historically, filtering approaches to calculating bridge displacement from acceleration signals generally look to treat the acceleration signal to remove or reduce the noise in the signal such that accurate displacements can be calculated. Baseline correction methods were demonstrated in [10,11], with high pass filters applied to deal with the presence of noise in accelerations. In both instances, the recovered dynamic displacements showed good agreement with the verification displacement measurements. Similar successes were displayed in Park et al. [12], using a velocity estimation method that also featured a high pass filter. Other approaches that have yielded positive results include [13], where a polynomial fitting approach was applied to the acceleration signal before estimating displacement, and [14], where a state space model representation of acceleration was used to calculate displacements. The references above are not exhaustive but the approaches the authors propose are characteristic of the traditional filtering approach to the problem of noise in bridge response acceleration signals.

The past few years have seen further evolutions of filtering techniques being developed. For example, in [14,15], a displacement reconstruction approach, developed in [16,17], was used to measure vertical and transverse displacements from a timber railway bridge using a finite impulse response (FIR) filter. Similarly, a Fast Fourier Transform - Direct Digital Integration (FFT-DDI) technique was demonstrated in [18,19] to calculate displacements from a steel railway bridge. In a data fusion approach in [20,21], a Kalman filter was applied to time synchronised strain and acceleration measurements to correct the signal drift in displacements calculated from accelerations due to a truck crossing a bridge. Each of these filter-based approaches suffer from the same issue of sensitivity to parameter selection, i.e., the methods work well provided appropriate filtering thresholds/parameters are used in a particular scenario. However, without having a verification displacement it is not possible to know if the correct thresholds are being used and/or if the correct displacement is being returned. Therefore, these approaches have not typically been used on other bridges without a verification displacement such as from a camera or LVDT.

A review of displacement estimation approaches was carried out in Arias-Lara and De-la-Colina [22] that featured seven separate methods for calculating displacement from acceleration for civil structures. The study considered the merit of each approach for application in different situations, concluding that there was no one preferable solution to displacement estimation to address all scenarios. The two methods that achieved the most accurate displacements for two bridge types were only accurate on one bridge type each, showing a lack of robustness with many displacement estimation approaches.

1.3. Improved hardware in displacement estimation

Although still using a filtering approach, a simpler to implement procedure was proposed in Sekiya et al. [23] and tested on nine different accelerometers of varying quality. A combination of bandpass and lowpass filters were used to remove low frequency noise from acceleration signals before calculating displacement from short duration truck loading on a concrete bridge. The study highlighted the improved accuracy of calculated displacements relative to the quality of the accelerometer used. This becomes a noise minimisation problem, that is, selecting appropriate accelerometers to reduce noise in acceleration signals, though still filtering the signals to remove a reduced amount of noise. While seeking to minimize errors in calculated displacements for the FFT-DDI method, the authors in Ribeiro et al. [24] acknowledged that where their error estimation approach does not work, improved hardware with lower noise could be a solution.

In Hester et al. [25], the authors proposed a method for calculating displacement from unfiltered accelerations with an inbuilt quality check. A series of accelerometers of varying quality were first tested in a lab to assess the levels of noise in the acceleration signals. To minimise noise, the highest quality accelerometer was then selected to measure midspan accelerations from a concrete

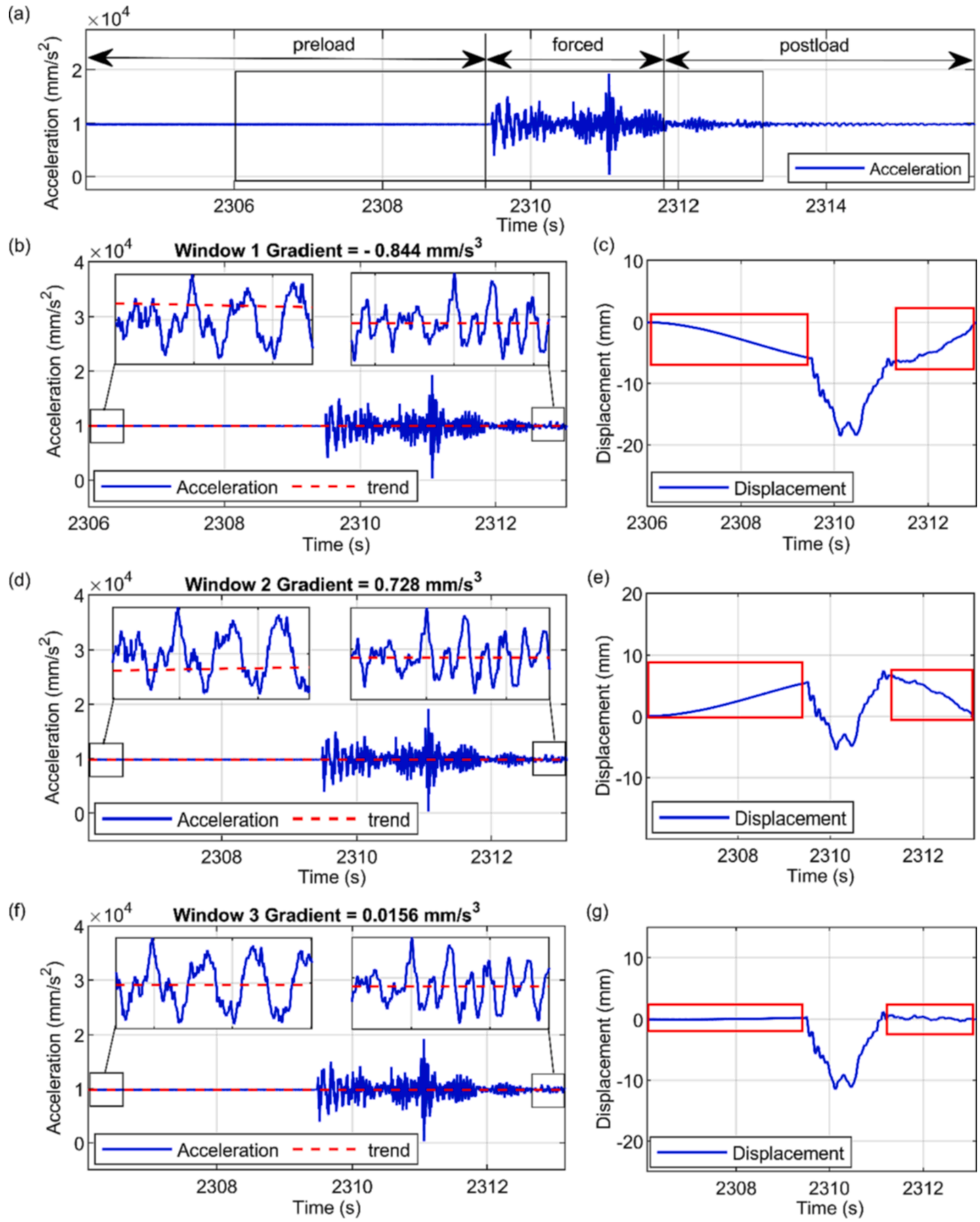

Fig. 1. Schematic of proposed approach trialling different acceleration start and end points, (a) full raw acceleration showing preload, post load and forced portions of the signal, (b) Window 1 raw acceleration (blue) and trend (red dashed) with start and ends shown to larger scales, (c) calculated displacements for Window 1, (d) Window 2 raw acceleration (blue) and trend (red dashed) with start and ends shown to larger scales, (e) calculated displacements for Window 2, (f) Window 3 raw acceleration (blue) and trend (red dashed) with start and ends shown to larger scales, (g) calculated displacements for Window 3. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

bride due to a passing truck. Knowledge that the calculated displacement should be 0 mm and relatively flat for unloaded portions of the calculated displacement signal (before the truck arrives and after the truck leaves) was used to quality check the resulting displacements. Whilst accurate displacements were achieved, to date, the procedure has only been trialled on one bridge and for short duration moving loads of less than 10 s of acceleration data. The approach was sensitive to start and end points, which was not discussed previously. As the quality check is carried out visually after calculating displacements, the trial and error to find a suitable window of displacement can be challenging and time intensive to carry out.

1.4. Contributions in this paper

The research presented in this paper adopts a similar philosophy to [25] and uses appropriate hardware to minimise low frequency noise in acceleration signals and does not rely on filtering. However, it investigates a number of previously unexplored issues, and has the following specific contributions:

- The sensitivity of calculated displacements to errors in the acceleration signal detrend is demonstrated. A robust, exhaustive search technique is implemented that trials different start and end points with a linear detrend to identify optimal windows of acceleration data (to detrend) to calculate accurate displacements.
- A novel adaption to the above approach is introduced that exploits general train and bridge information to accurately calculate bridge displacement from acceleration responses due to longer duration train loading.
- The abilities and the limitations of the proposed approach are explored with calculating bridge displacement from acceleration signals from pseudo static loading scenarios (when a moving load stops on the bridge for a period of time).

Unlike in previous studies, where proposed approaches are demonstrated on a single bridge, in this paper the proposed technique is demonstrated to be robust across a range of different bridge and load types. Section 2 presents a summary of the proposed approach, and Section 3 demonstrates its application in a short duration highway loading scenario. Section 4 introduces modifications to the approach for train loading scenarios, and Section 5 demonstrates its application to a railway bridge. Section 6 presents further load test scenarios carried out on a concrete highway bridge with modifications to the approach introduced with each new load case.

2. Proposed approach in this paper

The accelerometers used in this study are navigation grade uniaxial force balance accelerometers (QA-750) manufactured by Honeywell. They provide excellent performance with respect to noise, $<0.000069\ \mathrm{m/s^2/\sqrt{Hz}}$ (0–10 Hz) and $<0.00069\ \mathrm{m/s^2/\sqrt{Hz}}$ (10—500 Hz) [26]. In Hester et al. [25] it was demonstrated that the signals from these accelerometers could be used to calculate bridge displacements for a load test. Windows of raw acceleration were detrended and twice integrated, and assuming zero initial velocity and displacements, a quality check was used to evaluate the quality of the calculated displacements. The integration procedure required windowing the acceleration data such that there was:

- a short portion of signal before the vehicle arrives on the bridge (preload),
- the forced part of the signal, where the vehicle is on/crossing the bridge, and
- a short portion of signal after the vehicle has departed the bridge (postload).

The quality check performed on each window checked that the preload and postload portions of the displacement were close to 0 mm as would be expected using engineering judgement, offering confidence in the (calculated) forced displacement. However, there are challenges in implementing the procedure as it relies on zero initial conditions and the calculated displacements are sensitive to the start points of the window, therefore the trial-and-error process to find the right window is time intensive. The approach presented in this paper requires trial and error to identify optimal windows of acceleration to calculate accurate displacement and utilises a similar windowing philosophy to Hester et al. [25] that allows quality checks to be used to identify the accuracy of trialled windows of data. However, where previous approaches use zero initial conditions to detrend accelerations signals, the approach in this paper utilises a linear detrend for each trialled window of raw acceleration to trial a range of detrends. Varying start and end points result in subtly different trends being observed for different windows of acceleration, where the calculated displacement is sensitive to incorrect detrends, and errors in detrended accelerations result in magnified errors in the calculated displacements. Therefore, quality checks are required to identify optimal windows of acceleration that result in optimal detrends being performed, resulting in accurate calculated displacements.

To demonstrate, Fig. 1 presents a short duration highway loading scenario for a bridge. Fig. 1(a) presents the raw acceleration data where the load event, and therefore preload and post load portions of the signal, can clearly be identified. From this window of data, a number of unique windows of acceleration can be cut around the load event that include a portion of preload and post load signal, and for each window cut, a linear trend can be fitted to the acceleration data.

The black box in Fig. 1(a) identifies one such window, Window 1 (2306.00 s: 2313.04 s), and this window of raw acceleration is repeated in Fig. 1(b). The raw acceleration is plotted in blue, and the trend identified for the signal is shown in red. The scaled view of the start and end of the acceleration signal shows the trend begins above the raw acceleration and declines with a gradient of −0.83 $\mathrm{mm/s^3}$. The result of removing this trend from the acceleration data and twice integrating the signal to calculate displacements is shown in Fig. 1(c). The results are immediately identifiable as inaccurate, as the preload and post load shoulders (red boxes) are neither flat nor close to 0 mm as one would expect.

Fig. 1(d) shows a second window of raw acceleration data (blue), Window 2 (2306.03 s: 2313.11 s), with the observed trend for the window (red). The scaled view of the start and end of this window shows the trend observed inclines, and this time begins below the raw accelerations with a gradient of 0.73 mm/s^3. The result of removing this trend and twice integrating the accelerations to calculate displacement is shown in Fig. 1(e). Again, the results are immediately identifiable as unreliable, as neither preload or post load shoulder remains flat or close to 0 mm.

Fig. 1(f) presents a third trialled window of raw acceleration data (blue), Window 3 (2306.10 s: 2313.14), with the trend observed for the window (red). The scaled view of the start and end point show that the trend observed has a subtle decline, with a gradient of 0.015 mm/s^3, and begins and ends slightly below the acceleration window start and end points. The result of removing this trend and twice integrating the accelerations to calculate displacements is shown in Fig. 1(g). The displacements for Window 3 appear to be accurate as both preload and post load shoulders remain flat and close to 0 mm for the duration of the shoulders (red boxes). This would suggest that the forced part of the signal would also be accurate.

The difference in trends observed between the three windows of acceleration, although subtle, result in errors in the detrended accelerations that are then amplified in the calculated displacements in Fig. 1(c) and (e). However, the errors are detectable in the calculated displacements with applied quality checks.

In order to deal with this sensitivity when calculating displacement from acceleration, start and end zones are first defined in the acceleration signal. The start and end zones are ranges of prospective start and end times for acceleration windows to be cut ahead of integrating. Each window, created from the range of start and end times, offers a small portion of preload and postload signal for shoulder evaluations and has a detrend unique to that window of acceleration. Fig. 2 shows the same acceleration data from Fig. 1, with start and end zones highlighted that would offer a diverse range of detrends to trial on the acceleration data. The extents of the forced part of the signal (i.e., when the truck is on the bridge) are marked with vertical lines, meaning even the shortest trialled window will have a short portion of preload and post load signal to quality check the calculated displacements.

The procedure for identifying the optimal window of acceleration to detrend and twice integrate to calculate displacements is described in the Flowchart in Fig. 3, with the first process being the definition of ranges of start and end zones in the raw acceleration signa (1). For each of the potential windows within the start and end zone boundaries (Fig. 2), the raw acceleration is cut and detrended, then integrated once to calculate velocity, then integrated a second time to calculate displacement (2). For each of the unique windows of (detrended) acceleration, quality checks are implemented to assess the resulting displacement signals. The first check that can be made in practical applications is using threshold displacement values to alert poor candidate windows of data (3). That is, significant positive displacements would be highly unlikely in most single span bridge scenarios, and this can be used as part of the quality checking processes where results that breach the threshold displacement are excluded from further consideration (6). The remaining quality checks implemented are more tailored for each bridge and loading scenario and result in quality indicators for each trialled window of data (4). For the moving truck crossing a bridge in Figs. 1 and 2, it is reasonable to use the preload and postload shoulders of the calculated displacements. The check here evaluates the preload and postload zones of the calculated velocity from the first integration of the acceleration data, where the velocity represents the gradient of the displacements (distance over time). The smallest gradient is indicative of the flattest preload and postload shoulders. When evaluating the flatness of displacements, it is important to ensure that the inclining and declining displacements do not cancel out. This is achieved by first summing the absolute velocities of these signal portions, before calculating the mean absolute value (by dividing by the number of velocity data points summed). Each window of data is assessed and sorted by the flatness of the preload and post load shoulders, representative of the likely most accurate window for the load event (5).

The window of calculated displacements presented in Fig. 1(g) was identified as the most likely accurate window using the procedure outlined in the flowchart in Fig. 3. However, a number windows could have been selected from within the two second start and end zones that would have yielded accurate calculated displacements. To demonstrate, Fig. 4 presents six (of over 250,0000 considered) windows of calculated displacements plotted in blue with camera measured displacements in red (the camera system used to measure the (validation) bridge displacement was the commercially available Imetrum Dynamic Measurement Station (DMS) system and further details are given later in Section 3.3). Fig. 4(a) is the top ranked window of calculated displacement, repeated from Fig. 1(g). Fig. 4(b)–(f) show five further windows that could be considered as estimating correct calculated displacements. For each window of calculated displacements, the start and end points fall within the range of start and end zones from Fig. 2. Collectively, Figs. 1 and 4 demonstrate that it is crucially important to identify a suitable window of acceleration (to detrend) as the calculated displacements are sensitive to the detrend applied to the raw acceleration data. The proposed approach is capable of trialling a range of

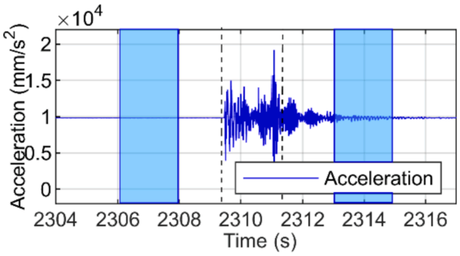

Fig. 2. Sample acceleration data with start and end zones highlighted for range of windows to be checked.

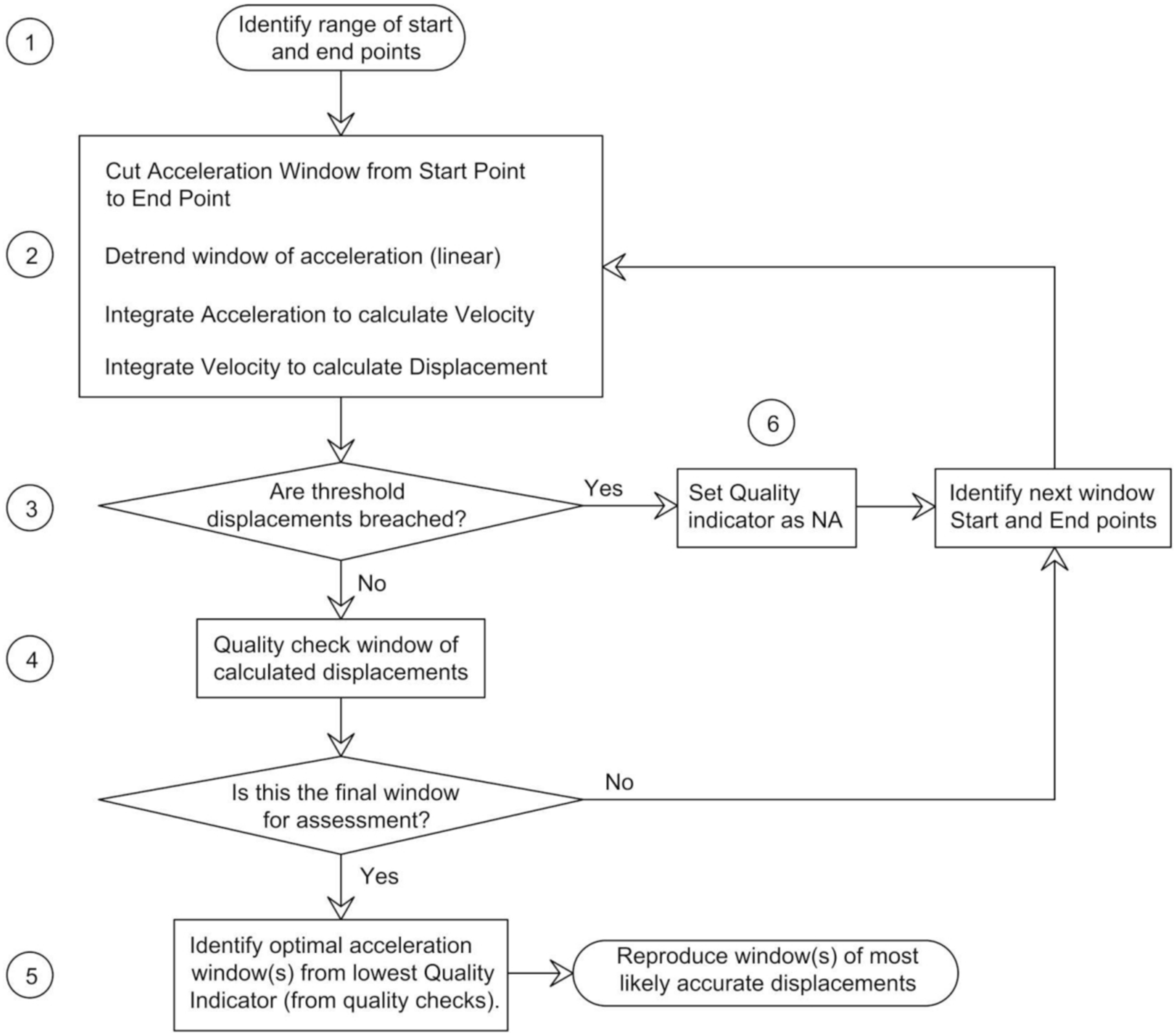

Fig. 3. Flowchart of proposed approach (general).

detrend solutions, influenced by varying start and end points of the signal, and using quality checks can identify an optimal window of acceleration, to detrend, to calculate accurate displacements.

The above is a singular application of the basic procedure. To demonstrate the repeatability and robustness of the approach across different bridges and loading types, Section 3 demonstrates the procedure as applied to a steel lifting bridge subject to truck loading. Section 4 presents modifications to the approach to accommodate railway loading scenarios. Section 5 demonstrates the modified procedure applied to a steel railway bridge subject to various train loads. Section 6 demonstrates the procedure applied to a highway bridge subject to a number of different loading scenarios including pseudo static loading, with further adaptions to the quality checking process to deal with the varying load scenarios.

3. Bridge 1 – Steel lifting bridge subject to short duration highway loading

This Section examines the ability of the approach proposed in Section 2 to calculate the midspan displacement of a single span steel lifting bridge under short duration highway loading. Details of the bridge and vehicle used are given in Sections 3.1 and 3.2. The test set up and results are presented in Sections 3.3 and 3.4, respectively.

3.1. Bridge description

The bridge used in the first test is a 17 m span steel lifting bridge, carrying two lanes of traffic as well as a footpath. Although the bridge is a lifting bridge, when it is in its down position it is essentially a simply supported single span. The bridge has a ladder-deck

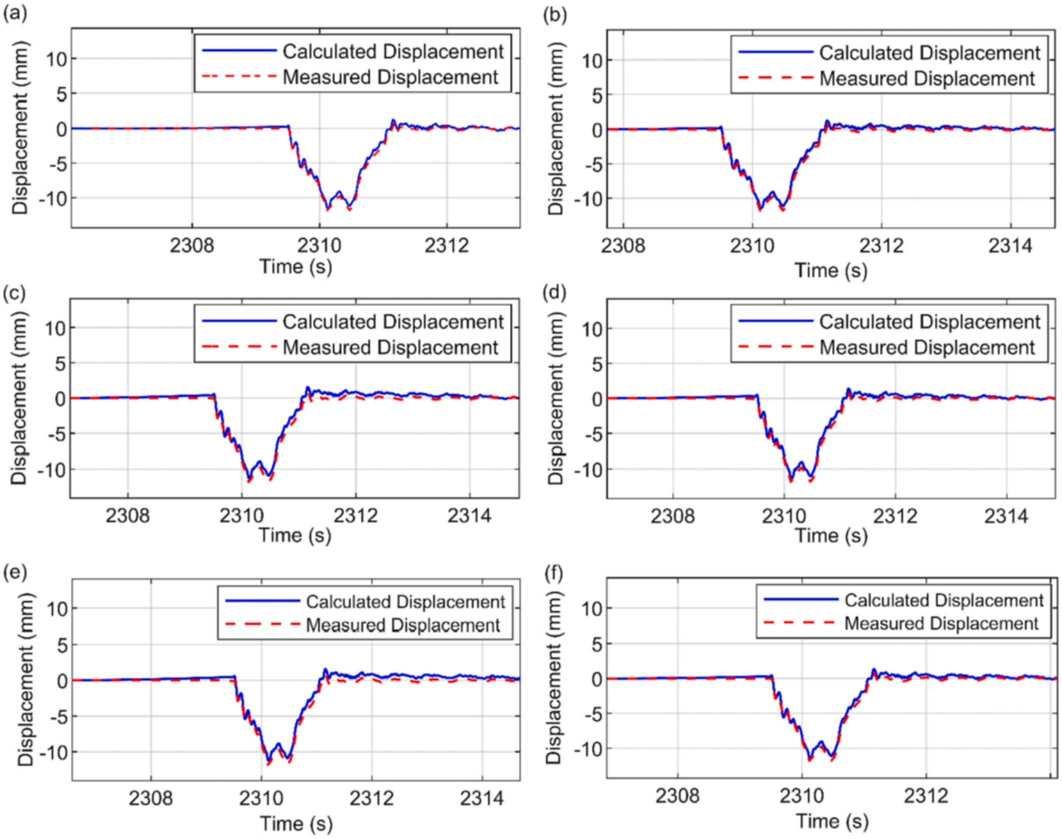

Fig. 4. Calculated displacements (blue) plotted with verification displacements (red), (a) top ranked window of calculated displacements, (b) - (f) alternative correct windows of calculated displacements. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

arrangement of longitudinal girders and lateral beams supporting a concrete deck. Fig. 5(a) is a photo of the west elevation of the bridge, and Fig. 5(b) is a photo of the bridge during testing, with the truck passing in the west lane of the bridge.

3.2. Vehicle used

The vehicle used in both highway bridge tests (Sections 3 and 6) was a Scania P450. Fig. 6 (a) is a photo of the truck used in the tests. The truck weighs 12 tonnes unloaded, and 32 tonnes when fully loaded. The axle loads for both the loaded and unloaded truck are provided in Fig. 6(b) and (c).

3.3. Test set up

For the steel lifting bridge test, the accelerometer was placed at midspan. The target for a displacement camera system was placed

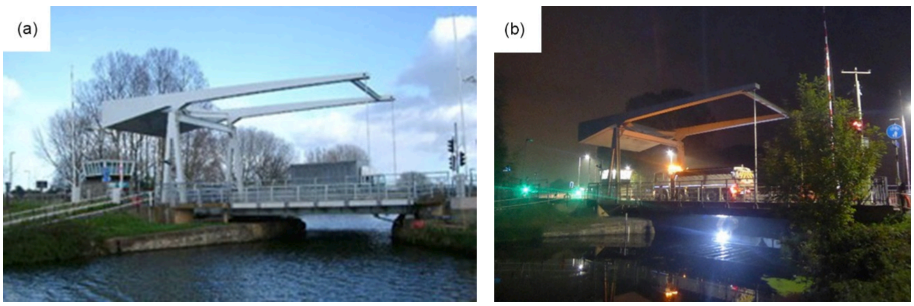

Fig. 5. Photo of Bridge 1, (a) west elevation, (b) west elevation with test truck passing in west lane.

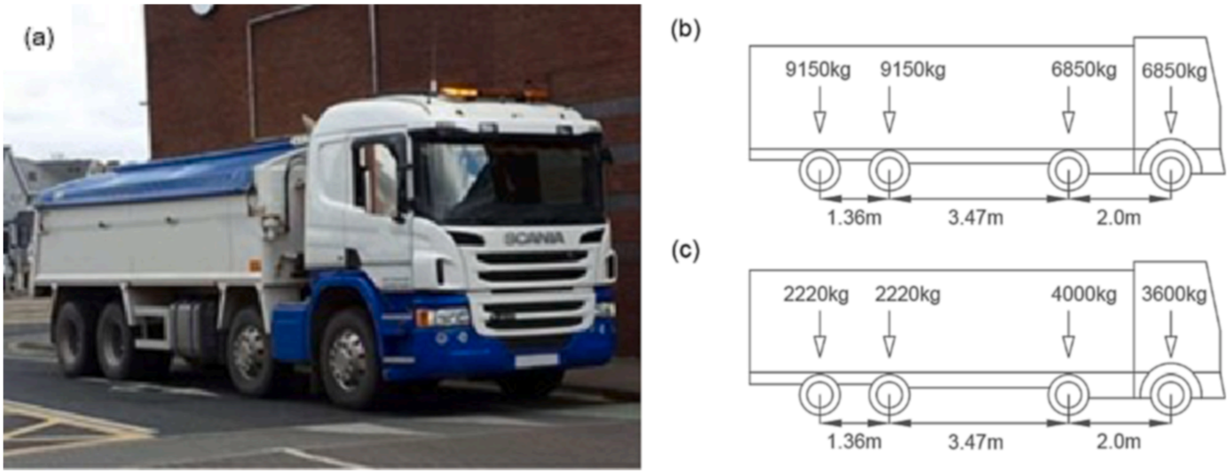

Fig. 6. Test truck used, (a) photo of the test truck, (b) loaded truck axle weights and (c) unloaded truck axle weights.

at the same longitudinal position. The Imetrum Dynamic Measurement Station (DMS), used to directly measure the bridge displacement, will be used to verify displacements calculated from integrating the accelerations. Fig. 7(a) shows a diagram of the bridge used and the test set up, highlighting the monitoring location, road lanes and direction of travel as well as annotating the support conditions. Fig. 7(b) is a photo of the Imetrum camera on the tripod and Fig. 7(c) is a photo of the west elevation of the bridge with the camera target and accelerometer location highlighted in red. Fig. 7(d) is a close-up view of the camera target mounted on the bridge and Fig. 7(e) is a photo of the accelerometer in perspex housing, installed on the bridge.

3.4. Results

A number of passes were made from the loaded truck for the steel lifting bridge load tests, recorded by both Imetrum DMS and QA accelerometer. The carriage lanes on the bridge are one way, and so the truck entered at the roller (south) end of the bridge and departed at the pinned (north) end of the bridge using both the eastern and western lane in alternating passes. The truck travelled at approximately 15mph for each of the passes, giving a 2.0–2.5 s loading duration. Fig. 8 shows the unfiltered midspan acceleration data for the tests plotted in blue against the left Y axis and the Imetrum camera recorded displacement (for verification) plotted in red against the right Y axis. The dashed boxes highlight the first two passes of the truck and magnified views are provided for detail. For the first pass of the truck (between 1,560 s and 1,570 s) the truck is travelling in the east lane of the bridge and the displacement of the west

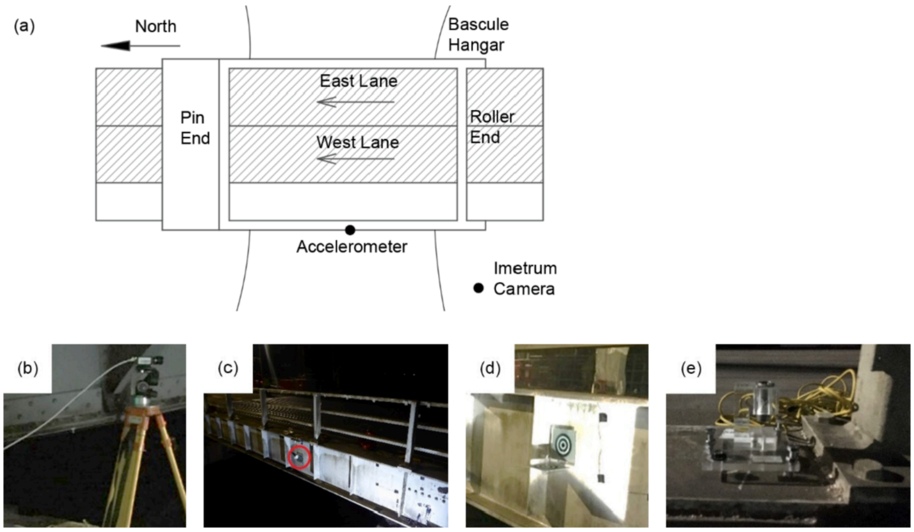

Fig. 7. Test set up, (a) schematic plan view of test set up with instrumentation locations, lane layout and supports annotated, (b) Imetrum camera on tripod, (c) west elevation of bridge with camera target and QA Accelerometer locations indicated with red circle, (d) close up view of camera target (e) close up view of accelerometer installed on the bridge. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

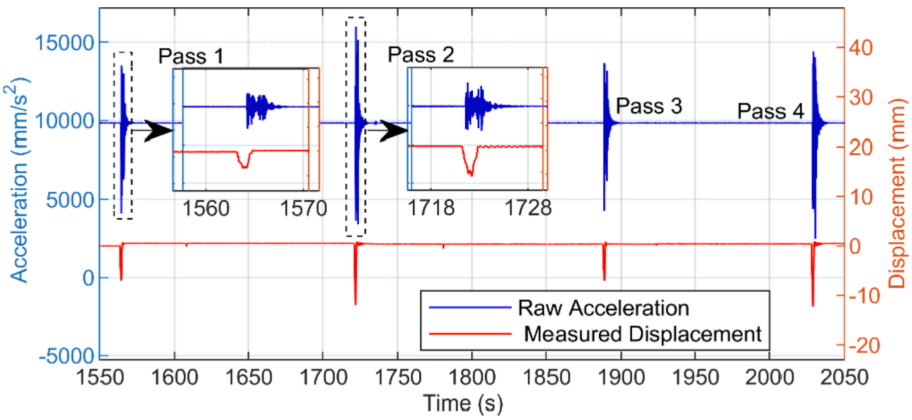

Fig. 8. Raw acceleration, plotted in blue to left Y axis, and measured displacement, plotted in red to right Y axis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

beam is approximately 5 mm. For the second pass (between 1,718 s and 1,728 s) the truck is travelling in the west lane and the displacement of the west beam is visibly larger at approximately 12 mm. The larger displacement observed in the second pass is simply because the truck is closer to the girder being monitored. For the third pass (1,890 s) and the fourth pass (2,040 s) the truck is in the east and west lanes respectively.

As this load test features a truck crossing the bridge in discreet, short duration loading events, the basic procedure outlined in Fig. 3 (Section 2) can be applied for each loading event, summarised below:

- Start and end zones (2 s each) are first defined around each load event. As the scanning frequency was 256 Hz, there are 512 possible start times and 512 possible end times giving ($512 \times 512 =$) 262,144 potential windows of acceleration.
- Each window is then integrated twice, and checked for:

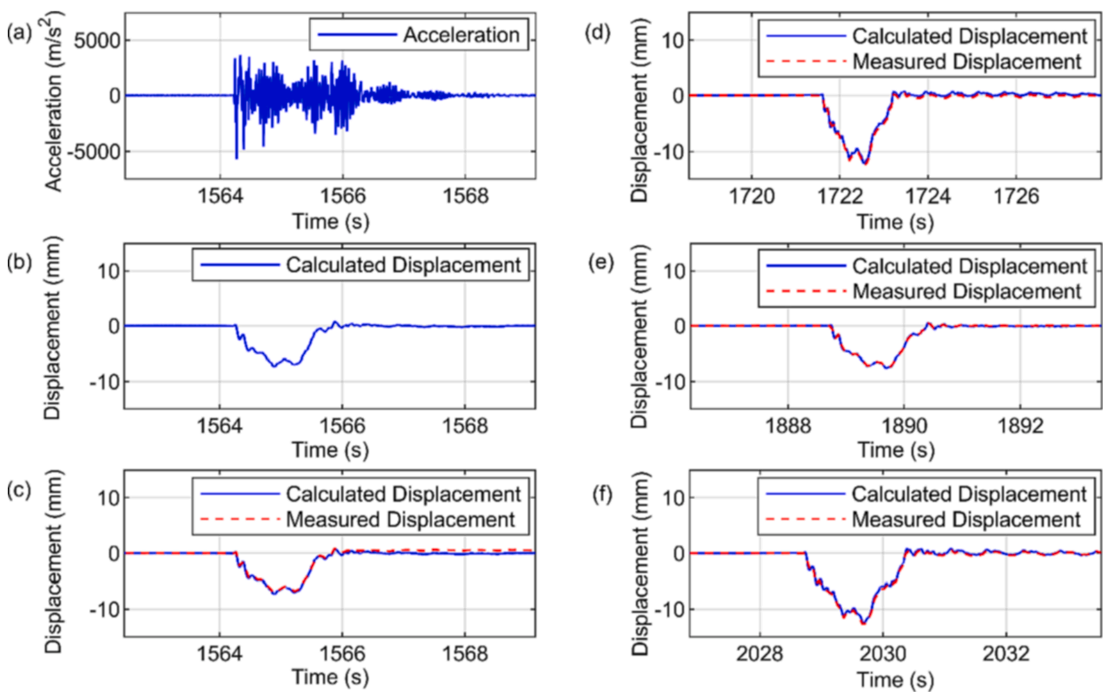

Fig. 9. Displacement calculated from acceleration, (a) top ranked acceleration window for Pass 1, (b) displacement calculated from Pass 1 acceleration, (c) Pass 1 calculated displacement & measured displacement, (d) top ranked window for Pass 2, calculated & measured displacement, (e) top ranked window for Pass 3, calculated & measured displacement, (f) top ranked window for Pass 4, calculated & measured displacement.

- o Threshold displacements, where positive displacements above 5 mm are unrealistic as the bridge is unlikely to lift off. A more precise threshold can be selected, however 5 mm improved processing time without requiring precise calculation.

  Windows that breach the threshold check are omitted from consideration, then for all the results that pass the threshold check,
- o The shoulder gradients are evaluated, assessing flatness (gradient ≈ zero) of preload and postload portions of the calculated displacements.

The 262,144 acceleration windows are then ranked based on the flatness of the preload and postload shoulders, where the flattest shoulders are indicative of the most likely accurate results. It is not necessary to calculate the gradient from the calculated displacement. Instead, it is determined by summing the absolute velocities (determined in the first integration) of the preload and postload signals in each window and dividing by the number of data points to calculate the mean gradient. As the displacement for this bridge is assumed to start at zero, the lowest mean absolute velocity is indicative of both; the flattest displacement gradient, and also the displacement that remains closest to 0 mm.

The data from the test was processed using MATLAB to carry out the above procedure. Fig. 9(a) shows the top ranked window of midspan acceleration for Pass 1, that is, when detrended and integrated twice, this window of data was found to offer the flattest preload and postload shoulders. This is demonstrated in Fig. 9(b) where, for the duration of the preload and postload portions of the signal, the displacement remains relatively flat and close to 0 mm. The flat shoulders offer confidence in the forced part of the displacement's accuracy. This confidence is found to be justified, as in Fig. 9(c) the calculated displacement is plotted alongside the verification displacement (measured by an Imetrum DMS) and shows good agreement. Fig. 9(d)–(f) show the top ranked windows of calculated displacements plotted with the measured displacements for Passes 2–4, respectively. Each of the calculated displacements are corroborated by the measured displacements for the duration of their respective windows. Collectively Fig. 9 shows that for short duration highway loading (less than 10 s) on a single span bridge, the proposed integration method is consistently giving accurate estimated displacements from the measured acceleration data.

4. Modification to approach for train loading scenario

Unlike with the first bridge test, where there was a single truck crossing the bridge, in this test, the trains consisted of a locomotive with various numbers of carriages. This led to increased loading durations, meaning that calculating displacements from acceleration becomes be more susceptible to corruption by low frequency noise. The change in scenario also results in a different displacement/time history shape being anticipated that can be used to assess calculated displacement accuracies. Instead of the single parabolic deflected shape observed in Section 3, for each train, there is effectively a series of parabolas as each set of axles crosses the bridge (as one set of axles is leaving, another is entering the bridge). For ease of discussion, we will refer to the parabolic forced displacements as troughs and the periods between forced displacements as peaks.

The spacing between the peaks and troughs is relative to the sensor location, bridge span and wheelbase of the locomotive/carriages. Fig. 10 presents a railway bridge scenario and the anticipated midspan displacement shape from the train crossing. Fig. 10(a) shows the loading scenario, where the bridge span is a similar length to the wheelbase of the train. As the length between sets of axles of the train and bridge span are similar, as one set of axles leave the bridge another set is about to enter. This means that the displacement signal will effectively be a set of consecutive peaks and troughs as shown in Fig. 10(b). Assuming similar loading per axle set, the peaks (P1-P5) return to 0 mm as the bridge is effectively unloaded between axle sets. The observation that the peaks will be approximately equal in height has the underlying assumption that the weight of the individual bogies will be approximately equal (or close to), which may not be universally true. However, in many instances, cargo wagons are approximately equal weight and passengers tend to distribute equally between carriages, which means that this assumption is often applicable.

The approximate prediction of the shape of the midspan displacement signal can be exploited to evaluate the likely accuracy of the calculated displacement due to a train crossing the bridge. This is carried out similar to the way that the preload and postload portions of the displacement signals in Sections 2 and 3 were used to determine the likely accuracy of the calculated displacement signal. Fig. 11 shows how the general information from a train and bridge can be used to search for peaks in the displacements from the train crossing. Specifically, it shows a schematic of a D6575 locomotive and two carriages, with the test bridge span either side of the front and rear wheels for reference. Dimensions have been included to show the bridge span, *B*, as well as the sensor location at the bridge midspan, *B/2*. The midpoints between consecutive sets of axles, or more specifically the midpoints between the end of one set of axles and the start of the next, are defined as M_i. These are indicated on the diagram and the distances to these points are indicated as m_i. Note the

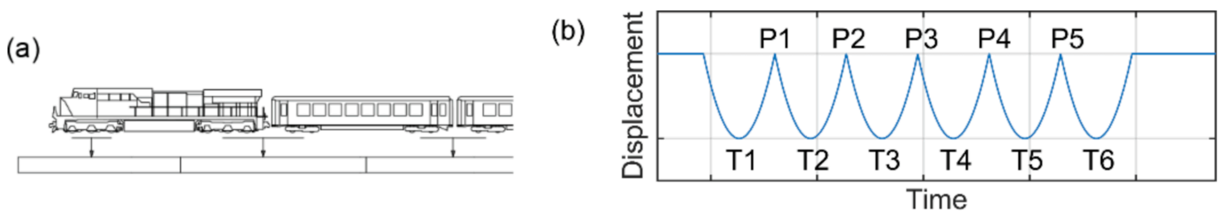

Fig. 10. Train loading scenario, (a) schematic of train with wheelbase spacing similar to the bridge span, (b) anticipated midspan displacement from scenario.

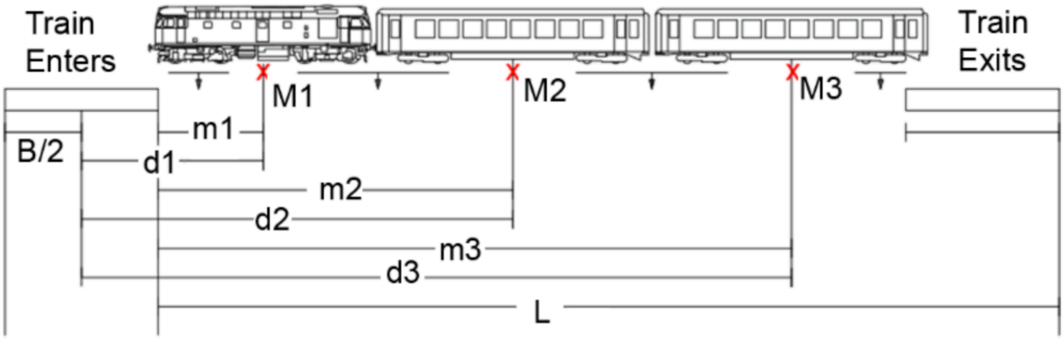

Fig. 11. Train 1 schematic, locomotive and two carriages with critical dimensions highlighted.

peaks, P_1-P_3, of the displacement signal occur approximately when these midpoints (i.e. M_1-M_3) pass the sensor location at mid-span of the bridge. Therefore, the distance from a given midpoint to the sensor location is d_i, where $d_i = m_i + B/2$. If the displacement is considered in the spatial domain, the distanced travelled by the front axle during the forced part of the displacement signal is essentially the entire wheelbase of the train plus the span of the bridge. This total distance is marked as L in Fig. 11. All the terms (B, Mi, m_i, d_i and L) shown in Fig. 11 can be estimated from general bridge and train geometry.

Assuming a constant velocity for the train, the forced displacement signal length (T) comes from the full train travelling a distance (L), and the peak positions (Pi) in the displacement signal can be estimated with Equation (1):

$$P_i = (d_i/L) \times T, \tag{1}$$

where T is the duration of time from the start of the signal to the end.

The precise peak locations can vary slightly; however, the approximated peak locations (Pi) allow the user to also approximate consistent spacing between peaks, and the base width of peaks. MATLAB's *findpeaks* [30] function was used to identify the actual peak locations and their magnitudes in the calculated displacements. The parameters selected for the function are:

- Minimum Peak Spacing – the difference in time between peaks,
- Minimum Peak Width – the difference in time between troughs (approximately similar to the minimum peak spacing in this instance),
- Minimum Peak Prominence – the difference in magnitudes between a given peak and the troughs immediately adjacent.

The parameters selected are used to identify the most prominent peaks in the displacement signal, that is, the peaks created when loaoded axle sets entering and exiting the bridge are repetitious and consistent peak patterns can be observed. The peak magnitudes and their corresponding timestamps are used to assess a relative gradient between the peaks, and the gradient is used with the preload and post load shoulder gradients as a quality indicator for each window of calculated displacements. As with the basic procedure (Fig. 3), the lowest gradient is indicative of the flattest shoulders, and most consistent peaks, and therefore most likely accurate window of calculated displacements.

5. Bridge 2 – Steel railway bridge subject to train loading

This Section demonstrates the application of the modified approach detailed in Section 4 to calculate the midspan displacement of a single span steel railway bridge under railway loading. Details of the bridge and vehicles used are given in Sections 5.1 and 5.2 respectively. The test set up is presented in Section 5.3, and 5.4 details the modified approach and results.

Fig. 12. South elevation of Bridge 2 – steel railway bridge.

5.1. Description of bridge

For the second series of tests, a steel railway bridge was used. The bridge has a span of 14.8 m and is skewed at an angle of approximately 60°. The concrete deck is supported by lateral steel beams at 1 m spacing, with ancillary steel bracing to the underside of the lateral beams. The lateral beams are fixed to two longitudinal girders that are simply supported on the abutments. Fig. 12 is a photo of the south elevation of the bridge.

5.2. Vehicles used

The bridge remained open and live trains were used for the load tests for this bridge. Photos of the three types of locomotives that featured in the test are shown in Fig. 13 where (a) is a D6575, (b) a 6960 Class 4-6-0 (Raveningham Hall), and (c) a 60103 Class 4-6-2 (Flying Scotsman). Each train had varying numbers of carriages, which created a range of loading durations from 10 to 40 s.

5.3. Test set up

The accelerometer and camera used in this test were the same as used in the previous test described in Section 3. The accelerometer was situated at the midspan of the bridge, on top of the northern longitudinal girder. The Imetrum camera was situated on the embankment north of the western abutment, facing southeast. A target for the Imetrum camera was placed at midspan of the northern longitudinal girder. Fig. 14(a) is a diagram of the test set up, showing the location of the Imetrum camera, accelerometer (and collocated target). Fig. 14(b) is a photo of the camera on a tripod (left side of image) with the longitudinal position of the camera target and accelerometer highlighted in red. Fig. 14(c) is a photo of the bridge deck, with the accelerometer location highlighted in red.

5.4. Modified approach applied to train 1

General information about the locomotive and carriages was sourced from the railway operator website to apply to the schematic in Fig. 11. The locomotive length, a D6575, was taken as 15 m from (front wheels to rear wheels) [28]. Most passenger cars on this railway are British Rail Mark 1 Vehicles [29] and are taken as approximately 19.5 m in length. To estimate the displacement peak positions in the acceleration signal, the entry and exit of the train need to first be identified in the accelerations. The entry of the train can be determined by the start of the excitation as the front wheels enter the bridge. The exit can similarly be estimated from the drop off in amplitude of the acceleration signal.

Fig. 15 (a) shows the raw accelerations from the first train crossing the bridge. The black vertical lines indicate the front wheels of the train entering the bridge (at 290.4 s) and rear wheels leaving the bridge (at 298.4 s). This gives the forced displacement duration for the test, $T = 8.0$ s. The bridge span is $B = 14.8$ m, and the train length (from front wheel to rear) is approximated as 54 m, with mid points between sets of axles occurring at 7.5 m, 34.5 m and 54 m from the front wheels of the locomotive. This gives the distance of the crossing as $L = 69$ m, and distance from the three midpoints to the sensor location (midspan of the bridge) as $d_1 = 15$ m, $d_2 = 34.5$ m and $d_3 = 54$ m. Using Equation (1), the three peak positions anticipated in the displacement signal can then be estimated as $P1 = 1.74$ s, $P2 = 4.0$ s and $P3 = 6.26$ s, from the train entry onto the bridge (290.4 s) and these are indicated in the accelerations in Fig. 15(a) with red crosses.

Whilst *P1*-*P3* were each estimated as being at least two seconds apart, the minimum peak width and spacing for the peak search was set to 1.5 s to allow for user error in the approximations. The actual peak prominence is unknown; although, it was sensible to estimate a minimum peak prominence of 1 mm to only identify the significant peaks in the signal without interference from smaller oscillations. This means that we would expect the difference in peak and trough magnitudes to be at least 1 mm. The bridge span is approximately equal to the axle set spacing of the train. Therefore, the peaks in the forced displacements should be of similar magnitudes, close to 0 mm, like in the loading scenario in Fig. 10. Similar to Section 3, we have a set of possible start points and a set of possible end points forming an array of unique acceleration windows. These start and end zones are two seconds long and the scanning frequency was 512 Hz, giving 1024 starting times and 1024 end times for a total of 1,048,576 possible windows. All windows of acceleration were integrated twice, and each calculated displacement then assessed for:

Fig. 13. Photos of locomotives that were used in Bridge 2 load test, (a) D6575 (Photo by Rob Hodgkins [27]), (b) 6960 Class 4-6-0 (Raveningham Hall), and (c) 60103 Class 4-6-2 (Flying Scotsman).

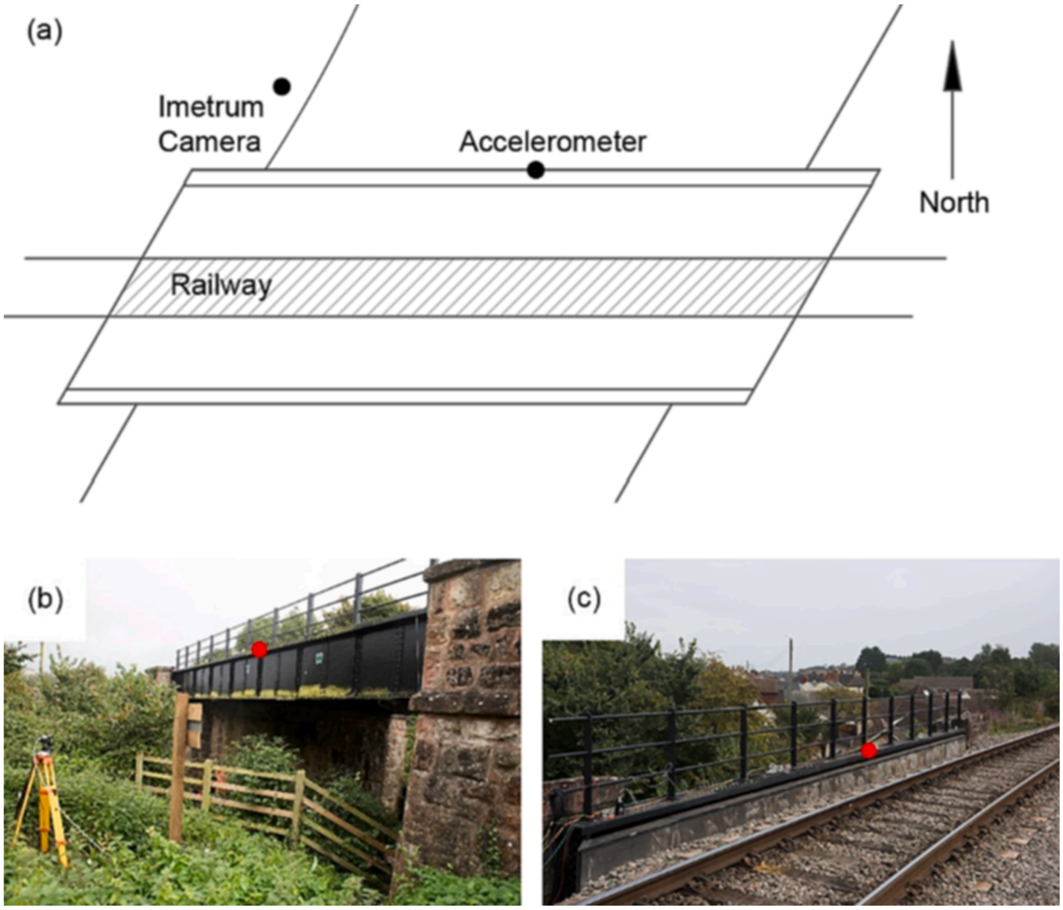

Fig. 14. Test set up, (a) plan view of Bridge 2 with monitoring equipment locations highlighted, (b) Imetrum camera on a tripod north of the bridge with the location of the camera target and accelerometer shown in red, and (c) top of the bridge with QA accelerometer location shown in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- Threshold displacements as with the basic procedure, where positive displacements above 5 mm are unrealistic as the bridge is unlikely to lift off. Windows that breach the threshold check are omitted from consideration.

Then for all the results that pass the threshold check,

- the gradient between the displacements at the three peak locations (peaks *P1-P3*), and
- The shoulder gradients, assessing flatness (gradient ≈ zero) of preload and postload portions of the calculated displacements.

Using the criteria above, the highest ranked displacement signal is shown in Fig. 15(b). The preload and postload shoulders have been highlighted with a red box to the left and right of the figure, respectively. The three interior peaks have also been highlighted with red crosses and labelled *P1-P3*. As well as being 0 mm for the duration of the preload and postload signal portions, the calculated bridge displacements are similar in magnitude (close to 0 mm) at *P1-P3*. This means one would have confidence that the amplitudes of the four troughs in the displacements are also correct. This confidence is justified as Fig. 15(c) shows a reproduction of the calculated signal from Fig. 15(b) and it matches well with the Imetrum DMS-measured displacement for the duration of the signal.

5.5. Results for trains 2–5

The train used in the demonstration in Section 5.4, Train 1 (pulled by a diesel electric locomotive), was selected as it had the simplest/most consistent axle geometry of all trains on the day. The remaining trains encountered featured either the 53808 (Class 2-8-0, Raveningham Hall) or 60103 (Class 4-6-2, Flying Scotsman) locomotives. The anticipated midspan displaced shape for the remaining locomotives/trains is slightly different to that of Train 1 as the first locomotive was the only to feature consistent spacing between all axle sets. That is, the multiple closely spaced wheels of the remaining locomotives and tinder wagons essentially result in a (close-to) uniformly distributed load at the start of the train. Fig. 16 replicates the train and bridge information given in Fig. 11 for the

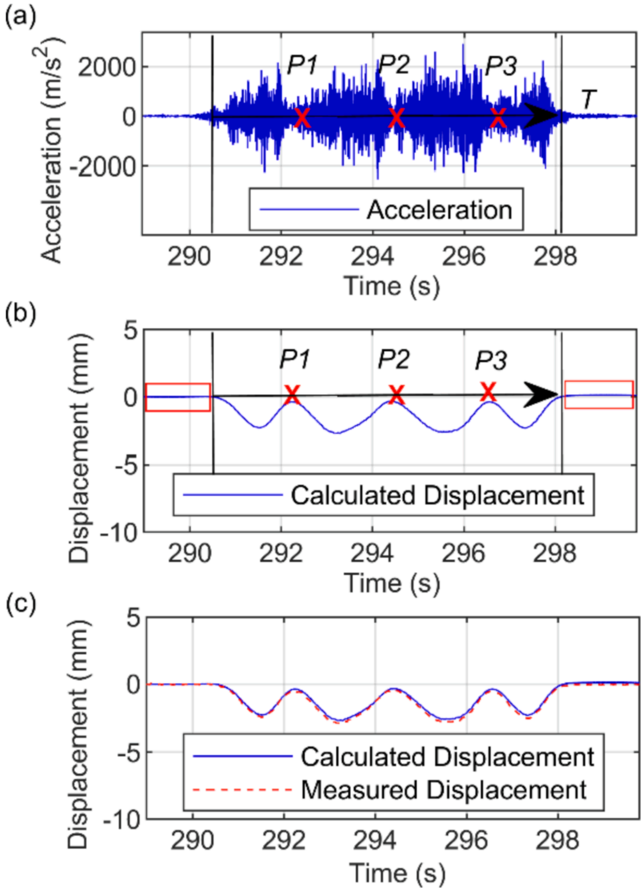

Fig. 15. Displacement calculated from acceleration for Train 1, (a) raw acceleration from train crossing the test bridge showing loading window (black vertical lines) and displacement peak locations P1-P3 (red crosses), (b) displacement calculated from accelerations highlighting preload and postload shoulders (red boxes) and peak locations P1 – P3, (c) calculated displacement (blue) and measured displacement (red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

two remaining locomotives. Fig. 16(a) shows the 5380 Class 2-8-0 locomotive (Raveningham Hall) and first two carriages. This schematic is relevant for Trains 2 and 3. Fig. 16(b) shows the 60103 Class 4-6-2 locomotive (Flying Scotsman) and first two carriages (this schematic is relevant for Trains 4 and 5). The classes refer to wheel configurations of the locomotives, where the 2-8-0 and 4-6-2 are the numbers of wheels at the front, middle and rear of the locomotives (excluding tinder wagons). The continuous wheels of the locomotives and then tinder wagons for these trains will cause a prolonged trough for the first forced displacement (when the locomotive is crossing the bridge). This means peaks do not form in the displacements until the carriages (which have much larger axle spacing) start to cross the bridge, and the distance between sets of axles becomes similar to the bridge span. With this consideration (i.e., number of peaks = number of carriages), Equation (1) estimates the peak positions P_1-P_n, with the positions of $M_1 - M_n$, etc starting from the first carriage of each of the remaining trains. The estimated peak positions then allow for defining of the minimum peak width and spacing for peak searches in the calculated displacements.

Table 1 gives general details for the four trains used for the remaining bridge tests, including locomotive type and number of

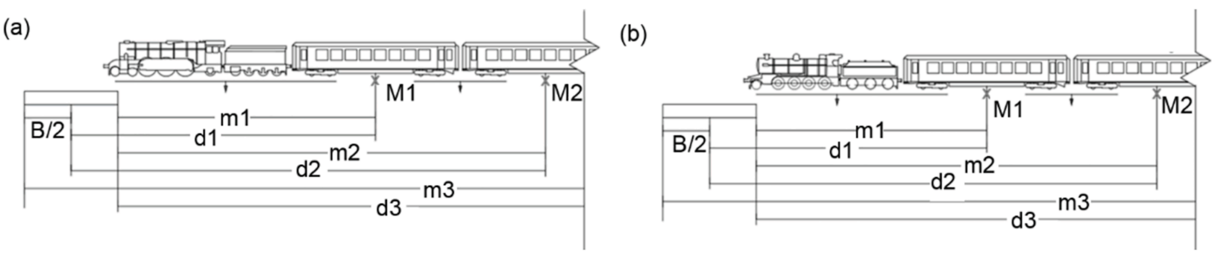

Fig. 16. Displacement peak estimation schematic, (a) 53808 Class 2-8-0 (Raveningham Hall), (b) 60103 Class 4-6-2 (Flying Scotsman).

carriages in the train. For each of the remaining trains, the estimated peaks were at least two seconds apart. As the displacement peaks would be resultant of carriage midpoints passing the sensor location, they would also be expected to be of similar magnitude to those from Train 1. Therefore, for simplicity, the peak search criteria for the remaining trains were kept consistent to that of Train 1, with minimum peak width and spacing of 1.5 s and minimum peak prominence of 1 mm. It was found that the method was not particularly sensitive to the value of minimum peak width/spacing used in the algorithm. This is because the primary function of these values is to prevent small oscillations in the displacements being identified as the peaks of interest.

For each train, two second start and end zones were again considered, searching for suitable windows of acceleration for calculating the midspan displacement. As the spacing between sets of axles of the carriages is similar to the bridge span, as was the case in Fig. 15, the peak magnitudes should be consistent and close to 0 mm. Fig. 17(a) shows the top ranked window of acceleration for Train 2. Fig. 17(b) shows the displacement for Train 2, calculated from the accelerations in (a). For the duration of the preload and postload portions of the signal, the displacements remain flat and close to 0 mm. Each of the peaks between troughs in the displacement return to a similar magnitude; close to 0 mm (with little difference between), which is what would be expected for this loading scenario (Fig. 10). Collectively these checks indicate that the accuracy of the calculated displacements is likely to be high. This is verified in Fig. 17(c) with the calculated and (Imetrum) measured displacements plotted in blue and red, respectively. Both measurement methods have good agreement for the duration of the signal. Fig. 17(d)–(f) shows the top ranked calculated displacements (blue) and verification displacements (red) for Trains 3, 4 and 5. Despite some small differences between the displacement methods at the carriage peaks, each window of displacement largely agrees with the verification displacement. Collectively Figs. 15 and 17 show the approach can cope with more complicated loading from trains for a short span railway bridge, irrespective of locomotive type, and is capable of delivering good performance consistently with only general train and bridge information required.

6. Bridge 3 – Concrete highway bridge subject to various highway loading scenarios

This Section examines the method's ability to cope with long duration static loading (i.e., parking the truck on the bridge), and if it remains performant for lower levels of load. Details of the bridge are provided in Section 6.1, and the test set up in 6.2. The results of the various tests are presented in Section 6.3.

6.1. Description of bridge

The bridge selected for the final tests is a 36 m span, simply supported bridge that carries two lanes of traffic, one in each direction. The bridge is a half through design, with two longitudinal steel girders and lateral steel beams supporting a concrete deck. Fig. 18 is a photo of the bridge's north elevation with the test truck passing close to the southern girder.

6.2. Test set up

The truck used in the third bridge tests was the same truck as used in the first bridge test, detailed in Section 3.2. For the final bridge tests, there were four types of tests carried out to assess the integration procedure's ability to cope under different loading scenarios. The tests featured both a loaded and unloaded truck, and load tests were carried out with a constant moving load as well as with the truck stopping at various positions on the bridge. Fig. 19 shows the test set up for the bridge. Fig. 19(a) shows a plan view of the bridge with the direction of travel, stopping locations (labelled A-F) and the instrumentation locations highlighted. Fig. 19(b) shows the QA accelerometer mounted to the underside of the top flange of the girder. Fig. 19(c) shows the monitoring station with Imetrum camera on the tripod. The instrumentation shown in Fig. 19(b)-(c) was on the south side of the bridge, and the north side had the same instrumentation.

The first series of tests was carried out using the loaded truck as used in Section 3. The truck crossed the bridge at a constant speed approximately 15-20mph without stopping, offering a forced displacement duration of 5–6 s. The second series of tests also used the loaded truck, but the truck came to a stop (for 10–20 s) at three stopping locations on the bridge before leaving the bridge. The stopping locations were at quarter, mid and three-quarter span; points A-C in the north lane and D-F in south lane, shown in Fig. 19(a)). These tests offered a forced displacement duration of around 90 s, the longest loading duration investigated in this paper. The truck was then unloaded for the remaining tests. For the third series of tests, the truck again crossed the bridge at a constant speed approximately 15-20mph without stopping. For the final tests, the unloaded truck stopped at midspan before moving off again, offering a displacement duration of around 40 s. Table 2 offers a summary of the test details for the results presented in Sections 6.3.1–6.3.4.

Table 1
Bridge test 2 - vehicles and peak search information.

Train #	Locomotive type	Locomotive length	Carriage length	No. of carriages	Train length	Duration (s)	Peak spacing
2	53808	17.9 m	19.5 m	8	163.9 m	27.7	1.5 s
3	Class 2-8-0					38.5	
4	60103	21.3 m			167.3 m	23.1	1.5 s
5	Class 4-6-2					27.2	

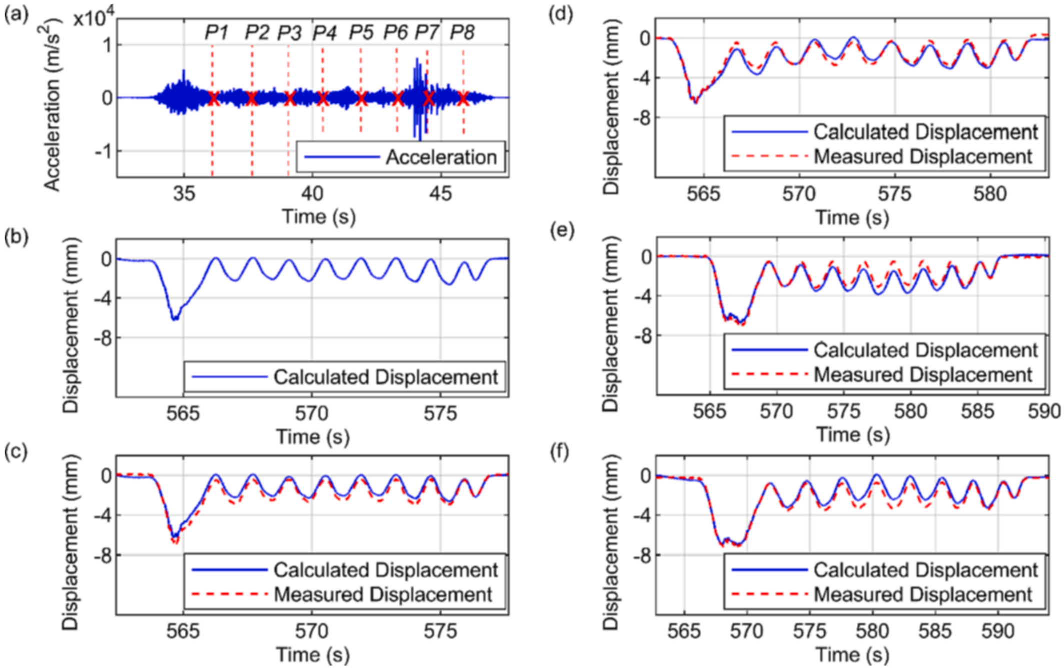

Fig. 17. Displacement calculated from acceleration for Trains 2 – 5, (a) top ranked acceleration window for Train 2, (b) displacement calculated from Train 2 acceleration, (c) Train 2 calculated displacement & measured displacement, (d) top ranked window for Train 3, calculated & measured displacement, (e) top ranked window for Train 4, calculated & measured displacement, (f) top ranked window for Train 5, calculated & measured displacement.

Fig. 18. North elevation of test Bridge 3 with test truck passing in the south lane.

6.3. Results

6.3.1. Loaded truck - constant speed

The first series of tests consisted of a loaded truck that crossed the bridge without stopping. The expected midspan displaced shape would be similar to that from the first bridge tests in Section 3, where smooth, flat displacement shoulders close to 0 mm would be good indicators of accurately calculated displacements. Therefore, the basic procedure as detailed in Section 2 (Fig. 3) and used in Section 3 is applied here. Fig. 20 shows the raw acceleration data (blue) plotted against the left Y axis with the Imetrum camera recorded displacement (red) plotted against the right Y axis, with all the measurements being recorded on the north side of the bridge. Several passes were made from the truck, which carried on forming a circuit to return across the bridge again in the opposite direction, and hence, lane. Despite other traffic being captured in the data, the truck passes are clearly identifiable in Fig. 20 for Passes 1 and 2 which have been highlighted with vertical dashed lines.

Two second start and end zones were selected for searching for accelerations windows to integrate as used in the previous Sections. Fig. 21 presents the results for Pass 1, with the top ranked windows of midspan acceleration for the north and south sides of the bridge shown in Fig. 21(a) and (d), respectively. The calculated displacements for the north and south sides of the bridge, Fig. 21 (b) and (e), both have flat preload and postload portions of the displacement at approximately 0 mm suggesting good accuracy in the forced displacements. The north side forced displacement (≈5mm) is appreciably smaller than the south side displacement (≈10 mm), which

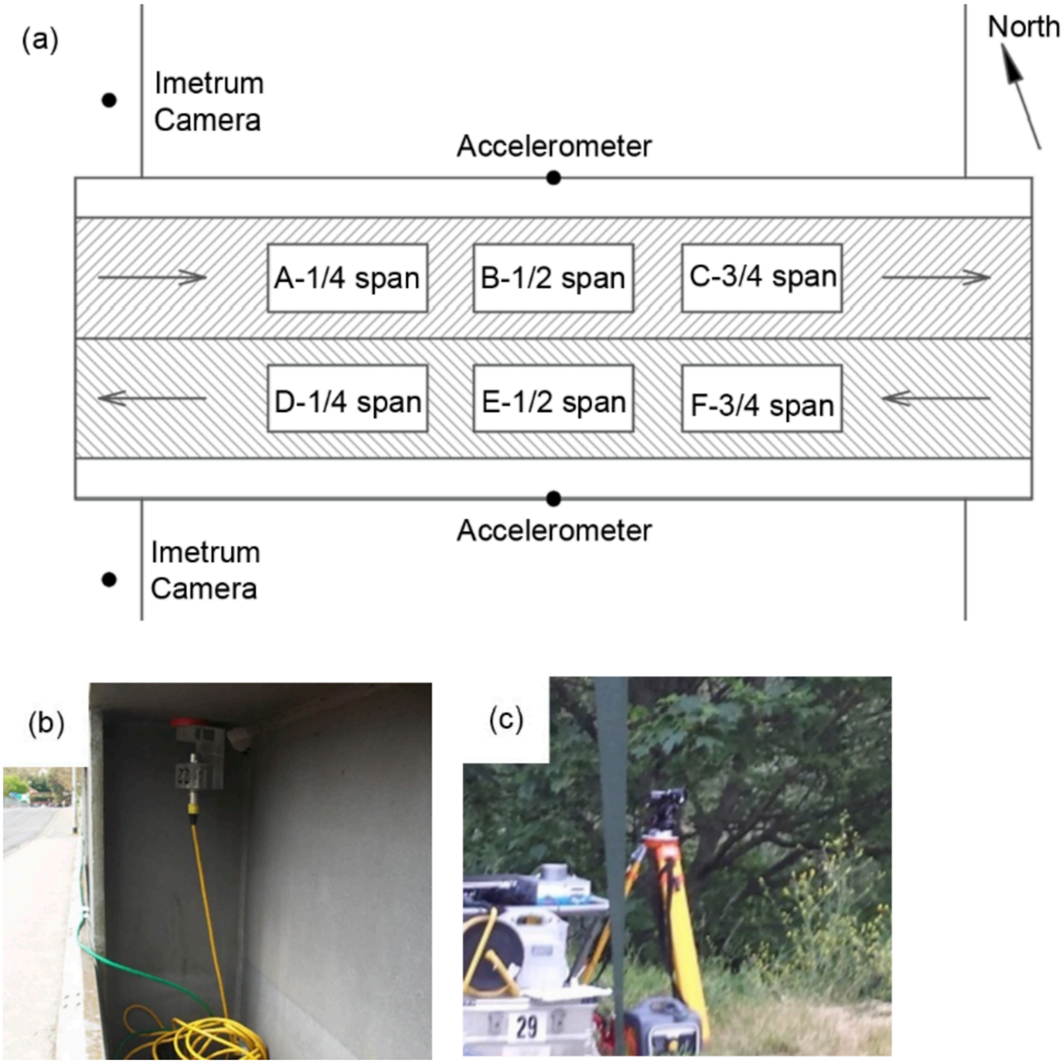

Fig. 19. Test set up, (a) Plan view of Station Road Bridge marked up with equipment locations, direction of travel, stopping positions and support conditions, (b) accelerometer attached to bridge, (c) Imetrum camera on tripod on western bank.

Table 2
Summary of test details.

Section	Test detail	Duration
6.3.1	Loaded Truck passing at constant speed of approximately 15-20mph	4–6 s
6.3.2	Loaded truck with stops at ¼, ½, and ¾ span positions, alternating between ABC and DEF, north and south respectively (Fig. 19)	80–90 s
6.3.3	Unloaded Truck passing at constant speed of approximately 15-20mph	5–7 s
6.3.4	Unloaded truck with stop at ½ span position, alternating between B and E, north and south respectively (Fig. 19)	30–40 s

is expected as the truck passed in the south lane. Fig. 21(c) and (f) show the calculated displacement in blue, and measured displacement in red, for the north and south sides of the bridge, with both windows of displacement showing near perfect agreement for their respective durations.

Fig. 22(a) and (b) show the calculated and measured displacements for Pass 2 north and south sides, respectively, where a near identical level of accuracy is observed between the measurement methods. The midspan displacements in Fig. 21(c) and (f), and Fig. 22 (a) and (b), collectively show that the procedure is not sensitive to bridge type. Further passes of the truck were carried out and with little variation in the quality of results obtained, where it was observed the method was repeatable.

6.3.2. *Loaded truck - three stops on the bridge*

The next series of tests consisted of a loaded truck that stopped at three positions when crossing the bridge, ABC in the north lane

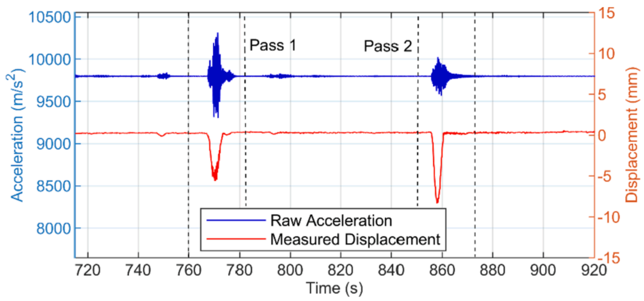

Fig. 20. Raw acceleration plotted in blue to left Y axis, and measured displacement plotted in red to right Y axis, for midspan of third bridge moving loaded truck test. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

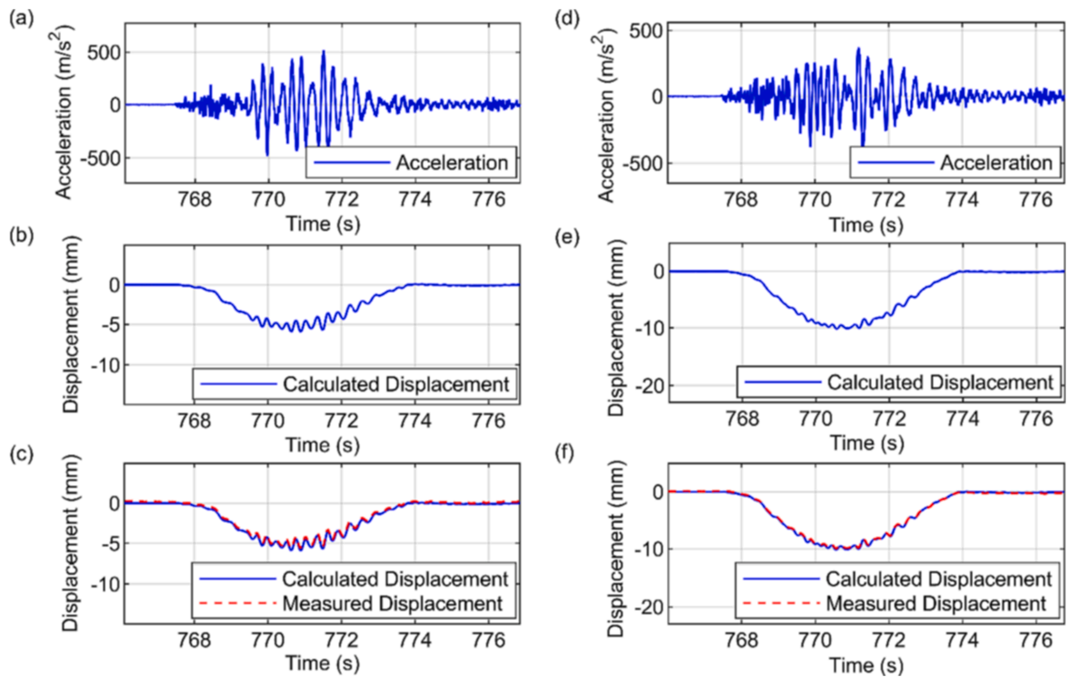

Fig. 21. Displacement calculated from acceleration for Pass 1 of loaded truck moving at constant speed, (a) top ranked acceleration window for Pass 1 north side, (b) displacement calculated from Pass 1 north side acceleration, (c) Pass 1 north side calculated displacement & measured displacement, (d) top ranked acceleration window for Pass 1 south side, (e) displacement calculated from Pass 1 south side acceleration, (f) Pass 1 south side calculated displacement & measured displacement.

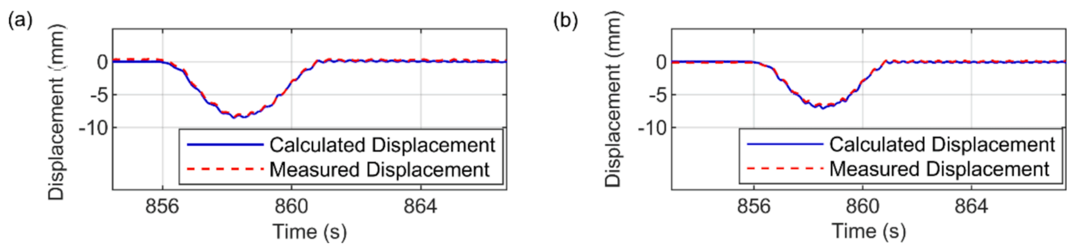

Fig. 22. Displacement calculated from acceleration for Pass 2 of loaded truck moving at constant speed, (a) Pass 2 north side calculated displacement & measured displacement, (b) Pass 2 south side calculated displacement & measured displacement.

and DEF in the south lane from Fig. 19(a). The expected displaced shape would therefore have three distinct periods where the midspan displacement should be relatively flat as the truck is stationary. Fig. 23 shows the raw acceleration data (blue) plotted to the left Y axis with the Imetrum camera recorded displacement (red) plotted to the right Y axis for two of the parked loaded truck tests. The

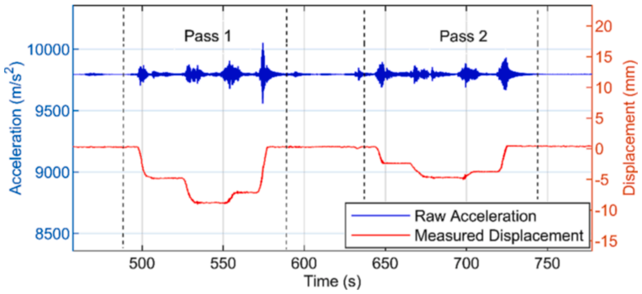

Fig. 23. Raw acceleration, plotted in blue to left Y axis, and measured displacement, plotted in red to right Y axis, for midspan of second bridge load test featuring three stops per crossing. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

acceleration and displacement shown in Fig. 23 were recorded on the south girder of the bridge and the stopping location of the truck is indicated on the figure. In each of the truck passes, three distinct flat periods can be seen where the truck stops at quarter span, mid span, and three-quarter span, before moving off the bridge. The loading events presented in this Section have been highlighted with vertical black lines, labelled with Pass 1 and 2.

The basic procedure as detailed in Section 2 was modified to assess the stationary-truck portions of the midspan displacements, with the preload and postload shoulders, for flatness. Similar to the preload and postload shoulder checks, this was carried out by evaluating the mean absolute velocity over the stationary-truck portions of displacement. For this series of tests, the initial threshold displacement check from the basic procedure (Fig. 3) had to be relaxed for some results to be obtained. The check omitted any data window from consideration that presented displacements beyond + 5 mm and −20 mm. The lower limit of −20 mm comes from an approximate calculation for one girder of the bridge and allowing some tolerance for errors, i.e., if all the weight of the truck was carried by one girder, a lower limit of −20 mm seemed sensible (also allowing for user error in estimates). Using these limits, however, meant that most of the candidate windows were automatically deemed unsuitable. Two second start and end zones were defined in the acceleration signals for each loading event, where optimal windows for calculating displacement were obtained. Largely, the procedure was unable to cope with such long duration loading, where the load durations exceeded 80 s.

Fig. 24(a) shows the raw acceleration from the north girder with the time periods when the truck was stationary at locations A, B and C indicated. The time periods/zones when the truck was stopped can be identified from the video footage recorded using a camera, an excerpt from which is shown in Fig. 19. Fig. 24(b) shows the north side midspan displacement for Pass 1, calculated from the accelerations in (a). It is immediately evident that the calculated displacements are unreliable as in each of the red highlighted zones (A, B and C) the displacements are not flat (i.e., even without seeing the verification displacement signal in Fig. 24(c), we know from engineering judgement that the value of the displacement time series should remain constant while the load is stationary). This lack of reliability is confirmed in Fig. 24(c) where the calculated displacement is plotted alongside the measured displacement and a poor match is observed. Specifically, the calculated displacement significantly overestimates the true displacement. Fig. 24(d)–(f) shows the same sequence for the same test but for data recorded on the south girder. The acceleration window identified in Fig. 24(d) was the top rated for the load test. The calculated displacements in Fig. 24(e) are not flat in the zones we would expect them to be flat and hence one would have little confidence in their accuracy. The suspected inaccuracy of the calculated displacement is confirmed in Fig. 24(f) where the calculated displacement (blue) is plotted alongside the measured displacement (red).

Fig. 25(a) and (b) present the calculated and measured displacements for Pass 2, north and south sides respectively, and again, the procedure was unable to cope with the duration of this test. The displacements in Fig. 25(b) show more promise than any of the other tests in this Section, in that there is some degree of flatness in the zones where one would expect the displacements to be flat. However, the overall shape of the calculated displacement is still such that one would have little confidence in the accuracy of the displacement. For each of the tests shown in Figs. 24 and 25, the results were inaccurate, and it is believed that the data window durations were too long due to the loading scenario for the approach to yield accurate displacements.

6.3.3. Unloaded truck - constant speed

The next series of tests involved using a constant moving unloaded truck for the bridge load tests. The testing procedure was the same as that used in Section 6.3.1 except this time the truck was unloaded, therefore the approximate axle weights are those shown in Fig. 6(c). The gross weight of the unloaded truck is approximately 12 tonnes resulting in displacements in the region of 2–4 mm, compared to the loaded truck (32 tonnes) used in Section 6.3.1.

The truck was travelling in the northern lane for the first pass (Fig. 19). Fig. 26(a) shows the midspan acceleration from the northern girder. The approach used to integrate the signal was the same as was used in Section 3 and Section 6.3.1. Specifically integrating 262,144 possible windows of acceleration and ranking them based on the gradient of the preload and postload parts of the calculated displacement. The calculated displacements shown in Fig. 26(b) was the highest ranked signal window and has flat shoulders, with subtle drift in the postload shoulder. Fig. 26(c) shows that the calculated signal was indeed accurate as it closely matches the measured displacement. Parts (d) - (f) of Fig. 26 present the same information but for the southern girder, i.e. the response of the southern girder for the first pass of the unloaded truck, where similar levels of accuracy are observed.

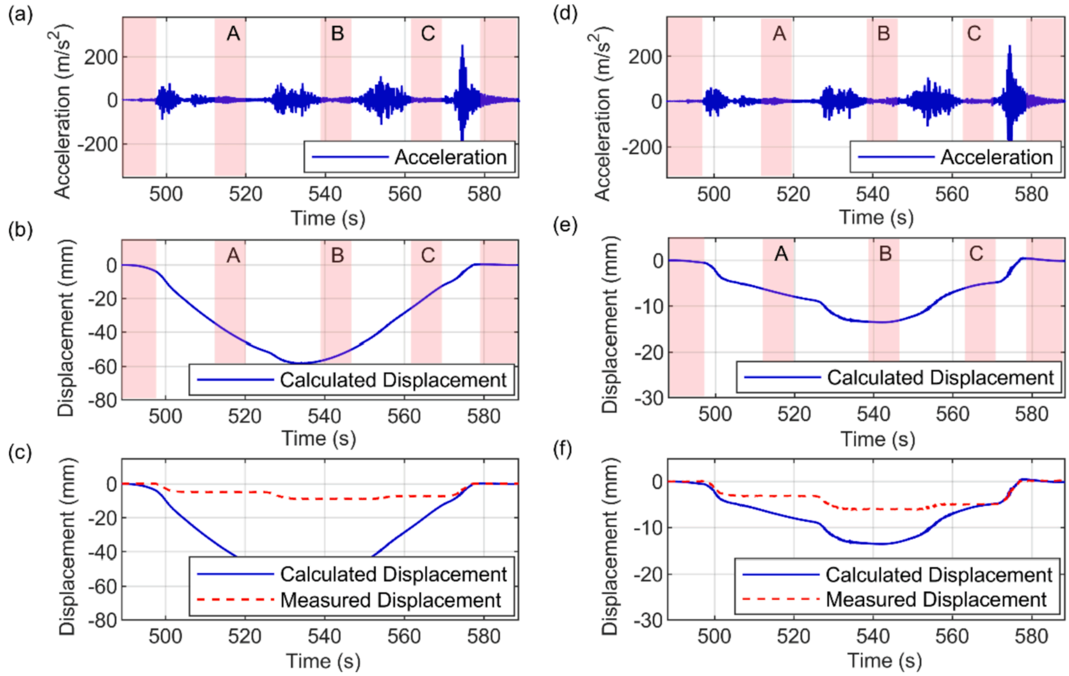

Fig. 24. Displacement calculated from acceleration for Pass 1 of loaded truck stopping at locations A-C (a) top ranked acceleration window for Pass 1 north side, (b) displacement calculated from Pass 1 north side acceleration, (c) Pass 1 north side calculated displacement & measured displacement, (d) top ranked acceleration window for Pass 1 south side, (e) displacement calculated from Pass 1 south side acceleration, (f) Pass 1 south side calculated displacement & measured displacement.

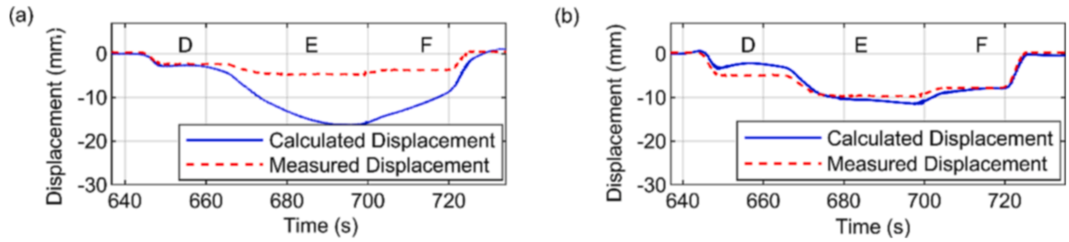

Fig. 25. Displacement calculated from acceleration for Pass 2 of loaded truck stopping at locations D-F, (a) Pass 2 north side calculated displacement & measured displacement, (b) Pass 2 south side calculated displacement & measured displacement.

Fig. 27(a) and (b) presents the calculated and measured displacements for the second pass of the unloaded truck, when the truck was travelling in the southern lane (see Fig. 19), for the north and south girders. Collectively, the repeated accuracy of the calculated displacements in Fig. 26(c) & (f) and Fig. 27(a) & (b) demonstrate that the approach is reliable even for small amplitude displacements, and therefore is not sensitive to load magnitude.

6.3.4. Unloaded truck - midspan stop on the bridge

The unloaded truck tests with stops on the bridge were subtly different to the loaded truck tests. Instead of stopping at quarter span, mid span and three-quarter span (ABC and DEF for north and south lanes, respectively in Fig. 19(a)), the truck only stopped at midspan (B and E) for a short period before departing the bridge. Stopping only once gave a shorter duration signal than stopping at three locations. To process the results, start and end zones were defined in the midspan acceleration signals for each loading event. Similar to the loaded truck with stops procedure from Section 6.3.2, the approach was modified to search a short portion of the mid-signal displacement for flatness (where the truck was parked at midspan).

Fig. 28 shows the results for the first pass of the truck when it was travelling in the south lane and stopped at point E shown in Fig. 19(a). Fig. 28(a) shows the top ranked window of acceleration, and Fig. 28(b) shows the calculated displacement for the north girder of the bridge. The three portions of expected flat displacement zones have been highlighted in red in Fig. 28(b). On an initial viewing the shape of the signal looks vaguely credible in the sense that the preload and postload parts are in the region of zero and the middle portion is staying relatively constant. However, on a more detailed inspection it is evident that the calculated displacement is not entirely flat in the places we would expect and hence one would have reservations about the accuracy of the predicted

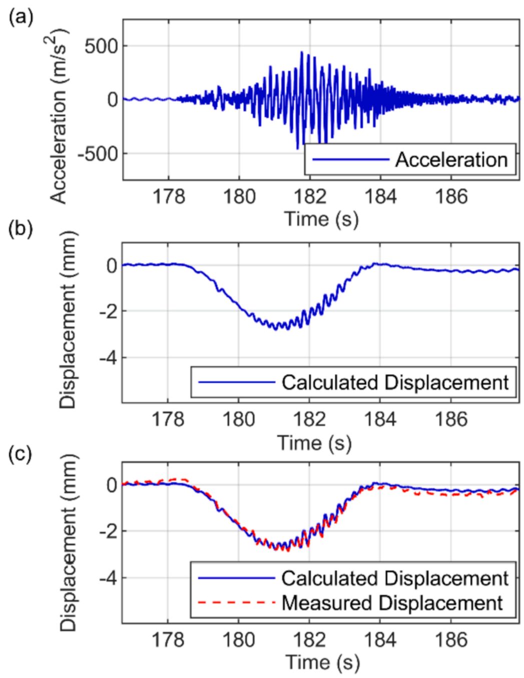

Fig. 26. Displacement calculated from acceleration for Pass 1 of unloaded truck moving at constant speed, (a) top ranked acceleration window for Pass 1 north side, (b) displacement calculated from Pass 1 north side acceleration, (c) Pass 1 north side calculated displacement & measured displacement, (d) top ranked acceleration window for Pass 1 south side, (e) displacement calculated from Pass 1 south side acceleration, (f) Pass 1 south side calculated displacement & measured displacement.

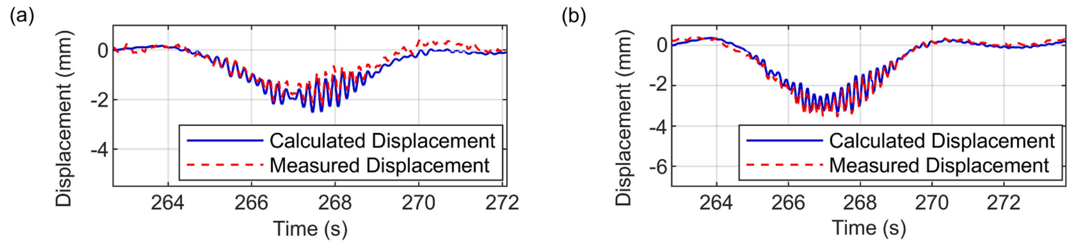

Fig. 27. Displacement calculated from acceleration for Pass 2 of unloaded truck moving at constant speed, (a) Pass 2 north side calculated displacement & measured displacement, (b) Pass 2 south side calculated displacement & measured displacement.

displacement. This is confirmed in Fig. 28(c) where the calculated displacement does not exactly match the measured displacement. Albeit the calculated displacement is noticeably closer to the measured value than it was for the longer duration static loading signals analysed in Figs. 24 and 25. Fig. 28(d)–(f) shows the same information but for the south side girder. Again, the calculated displacement, Fig. 28(e), is not entirely flat in the places we would expect and hence one would have reservations about the accuracy of the calculated displacement. The unreliability of the calculated displacements is confirmed in Fig. 28(f) when compared to the measured displacements.

Fig. 29(a) and (b) show the calculated and measured displacements (north and south girders, respectively) for the second pass of the truck, when it was travelling in the north lane and stopped at point B shown in Fig. 19(a). The calculated displacements in Fig. 29(b) seem reliable in the sense that they have (close-to) zero values and gradients in the places you would expect. Therefore, they are taken to be relatively accurate, and this is confirmed by the measured displacement.

As a general observation, the proposed integration approach performed appreciably better with the shorter duration pseudo-static loading in this Section than the longer duration static loading tests in Section 6.3.2. Admittedly only one of the calculated displacements could be considered correct (Fig. 29(b)). For the remaining three results (Fig. 28(c) & (f) and Fig. 29(a)) the magnitude of the calculated displacement has errors, but the broad shapes were reasonably accurate. This may indicate that the low frequency noise accumulated in the approximately 100 s long signals in Section 6.3.2 is too much for the method to cope with whereas the (approximately) 35 s signals occurring in is approaching a duration that the method could potentially cope with.

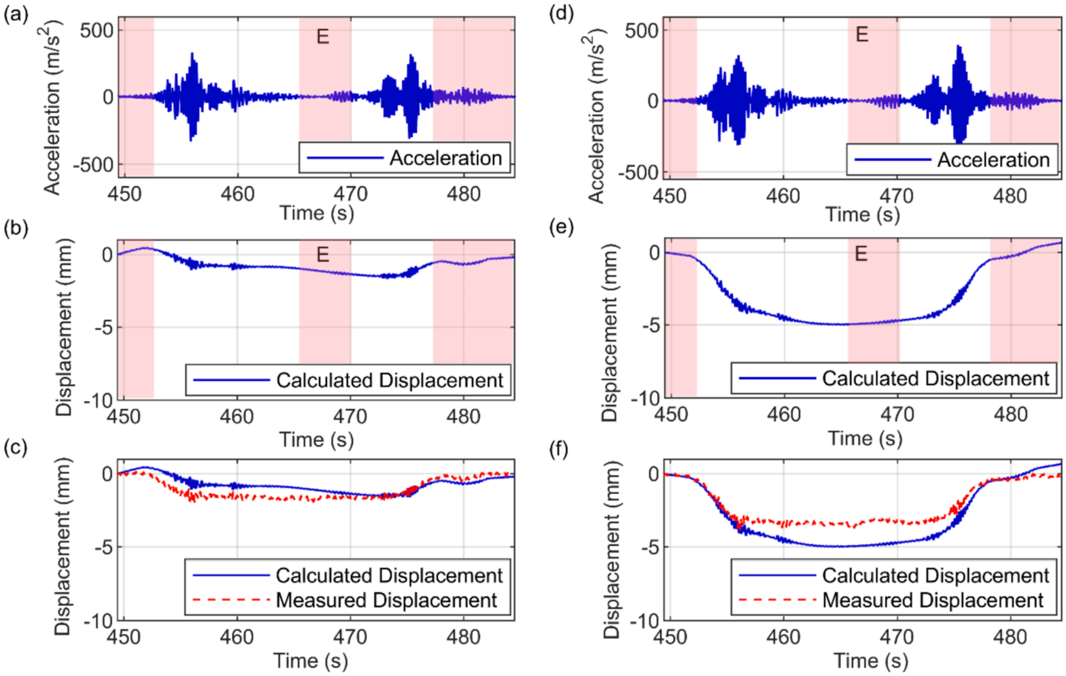

Fig. 28. Displacement calculated from acceleration for Pass 1 of loaded truck stopping at location E, (a) top ranked acceleration window for Pass 1 north side, (b) displacement calculated from Pass 1 north side acceleration, (c) Pass 1 north side calculated displacement & measured displacement, (d) top ranked acceleration window for Pass 1 south side, (e) displacement calculated from Pass 1 south side acceleration, (f) Pass 1 south side calculated displacement & measured displacement.

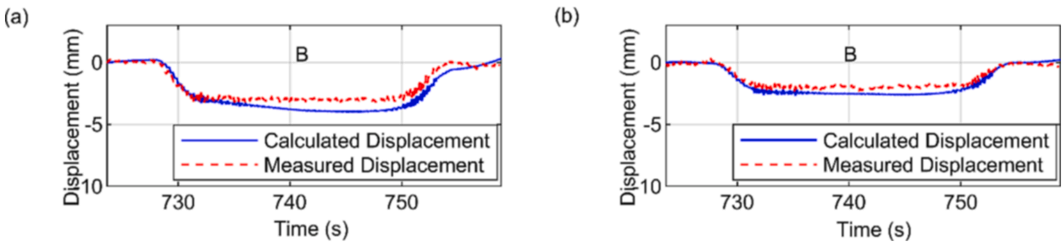

Fig. 29. Displacement calculated from acceleration for Pass 2 of loaded truck stopping at location B, (a) Pass 2 north side calculated displacement & measured displacement, (b) Pass 2 south side calculated displacement & measured displacement.

7. Conclusions

This paper proposes a robust integration approach to calculate bridge displacement from acceleration. Unlike many approaches proposed previously it does not rely on filtering to deal with the problem of low frequency noise corrupting the acceleration signal. Instead, it tries to cope with the issue of low frequency noise by (1) minimising the noise using appropriate hardware, and (2) integrating multiple windows of linearly detrended acceleration then exploiting anticipated displaced shape information (from the type of loading and the timing of the loading) to rank the various windows. Overall, the following observations can be made:

It is crucially important to select the optimal window of acceleration to integrate, as the detrend and calculated displacements are sensitive to this. Varying start and end points of the raw acceleration can be exploited to trial a range of detrends, where quality indicators can be used to identify the optimal window of calculated displacements.

The proposed approach works very well for short duration highway loading, as it has given very accurate results on two different bridges and for both loaded and unloaded trucks. Using general information about a bridge span and train (i.e., axle configuration), the procedure can also calculate accurate bridge displacements for railway loading scenarios. It was observed that as loading durations increase, the number of suitable candidate windows that result in accurate calculated displacements decreases.

The performance of the approach for static highway loading (i.e., trucks parked on the bridge) identified some limitations of the proposed approach. For the shorter duration static loading with signal length in the region of 35 s (Section 6.3.4) the method showed promise and in some instances was able to estimate displacements reasonably close the verification measurements. However, for the longer duration static loading of around 90 s (Section 6.3.2) the results were unreliable.

As with all approaches it will benefit from further testing e.g., other loading scenarios on other bridges with other hardware. However, overall, the approach appears robust and potentially useful.

CRediT authorship contribution statement

Andrew Bunce: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization. **David Hester:** Conceptualization, Methodology, Validation, Investigation, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Su Taylor:** Writing – review & editing. **James Brownjohn:** Validation, Investigation, Writing – review & editing, Funding acquisition. **Farhad Huseynov:** Investigation, Data curation, Writing – review & editing. **Yan Xu:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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